

Reduced Order Modeling (POD) for 2D Heat Equation

In this project, I solve a simple 2D heat equation using Python. I use a finite difference method to calculate the temperature over time. Then I use POD and SVD to reduce the amount of data.

Heat Equation: $dT/dt = \alpha * (d^2T/dx^2 + d^2T/dy^2)$

Numerical settings: grid = 50 x 50, dx = 1.0, dy = 1.0, dt = 0.1, and 100 time steps.

The temperature starts at zero. I put a hot square in the middle of the grid with temperature 1.0. The boundary temperature stays at zero.

Training parameters: alpha = 0.1, 0.5, and 1.0. I combine the snapshots from these values and build one common POD basis.

I use SVD to find the main patterns in the training data. I use the energy of the singular values to choose the number of POD modes.

The first 3 POD modes contain about 99.93% of the training energy.

Test parameter: alpha = 0.75. This value was not used during training. I use it to test the common POD basis.

The relative approximation error for the test parameter is 0.0068056, which is about 0.68%.

Conclusion: I used a simple POD approach for the 2D heat equation. Three POD modes keep about 99.93% of the training energy. The unseen test parameter alpha = 0.75 gives an approximation error of about 0.68%.